Lines, Triangles & Polygons

ANSWER KEY

Section 1: Coordinate Geometry of Lines

1 Find the perpendicular distance from the point $(3,\ -2)$ to the line $5x - 12y + 8 = 0$.

a. 2.5

b. 3.6 ✓

c. 4.0

d. 4.5

Formula: $d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}} = \frac{|5(3) - 12(-2) + 8|}{\sqrt{5^2 + (-12)^2}} = \frac{|15 + 24 + 8|}{13} = \frac{47}{13} \approx 3.6.$

2 What is the area of the triangle with vertices at $(0, 0), (6, 2),$ and $(4, 8)$ in the Cartesian plane?

a. 16

b. 20 ✓

c. 24

d. 30

Shoelace formula: $\text{Area} = \frac{1}{2}|0(2 - 8) + 6(8 - 0) + 4(0 - 2)| = \frac{1}{2}|48 - 8| = \frac{1}{2}(40) = 20.$

3 What is the acute angle of intersection between the lines $y = 2x + 1$ and $y = -3x + 4$?

a. 45° ✓

b. 60°

c. 30°

d. 75°

Formula: $\tan\theta = |\frac{m_1 - m_2}{1 + m_1 m_2}| = |\frac{2 - (-3)}{1 + 2(-3)}| = |\frac{5}{1 - 6}| = |\frac{5}{-5}| = 1 \Rightarrow \theta = 45^\circ.$

Section 2: Triangle Theorems & Inequalities

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4 A triangle has side lengths of 7 and 11. Which of the following CANNOT be the length of the third side?

a. 4 ✓

b. 5

c. 10

d. 17

By the Triangle Inequality: $11 - 7 < x < 11 + 7 \Rightarrow 4 < x < 18$. The length cannot be 4.

5 In $\triangle ABC$, $AB = 8, BC = 6,$ and angle $B = 90^\circ$. A line parallel to BC intersects AB at D and AC at E. If $AD = 6$, what is the length of segment DE?

a. 4.0

b. 4.5 ✓

c. 5.0

d. 5.5

By similar triangles $\triangle ADE \sim \triangle ABC$: $\frac{DE}{BC} = \frac{AD}{AB} \Rightarrow \frac{DE}{6} = \frac{6}{8} \Rightarrow DE = 6 \times \frac{3}{4} = 4.5$.

6 The side lengths of a triangle are $5, 12,$ and 13. What is the radius of the incircle (inradius)?

a. 1

b. 2 ✓

c. 3

d. 4

Inradius $r = \frac{A}{s}$. For a right triangle, Area $= \frac{5 \times 12}{2} = 30$. Semiperimeter $s = \frac{5 + 12 + 13}{2} = 15$. Thus $r = \frac{30}{15} = 2$.

Section 3: Polygons & Interior/Exterior Angles

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7 If the interior angle of a regular polygon is 150°, how many sides does the polygon have?

a. 10

b. 12 ✓

c. 15

d. 18

Each exterior angle $= 180^\circ - 150^\circ = 30^\circ$. Number of sides $n = \frac{360^\circ}{30^\circ} = 12$.

8 How many diagonals does a regular decagon (10-sided polygon) have?

a. 25

b. 35 ✓

c. 45

d. 70

Formula for diagonals: $\frac{n(n-3)}{2} = \frac{10(7)}{2} = 35$.

9 What is the area of a regular hexagon with side length 4 cm?

a. $24\sqrt{3}$ **cm**^{2} ✓

b. $36\sqrt{3}$ cm^2

c. 48 cm^2

d. $16\sqrt{3}$ cm^2

A regular hexagon comprises 6 equilateral triangles: $6 \times \frac{\sqrt{3}}{4}(4^2) = 6 \times 4\sqrt{3} = 24\sqrt{3}$ cm^2.

Section 4: Geometric Transformations & Dilations

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10 A triangle with an area of 14 cm^2 is dilated by a scale factor of $k = 3$. What is the area of the image triangle?

a. 42 cm^2

b. 84 cm^2

c. 126 **cm**^{2} ✓

d. 168 cm^2

Area scales by k^2: $14 \times 3^2 = 14 \times 9 = 126$ cm^2.

11 What is the matrix representation of a counterclockwise rotation of 90° about the origin?

a. $[[0, -1], [1, 0]]$ ✓

b. $[[0, 1], [-1, 0]]$

c. $[[1, 0], [0, -1]]$

d. $[[-1, 0], [0, 1]]$

Standard rotation matrix $[[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$. For $\theta = 90^\circ$: $[[0, -1], [1, 0]]$.

12 A line $3x - 4y = 7$ is translated 2 units right and 1 unit up. What is the equation of the new line?

a. $3x - 4y = 9$ ✓

b. $3x - 4y = 5$

c. $3x - 4y = 7$

d. $3x - 4y = 11$

Replace x with $x - 2$ and y with $y - 1$: $3(x - 2) - 4(y - 1) = 7 \Rightarrow 3x - 6 - 4y + 4 = 7 \Rightarrow 3x - 4y = 9$.

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